

MECH 462 Team Design 2026: Niagara Foot M3V01

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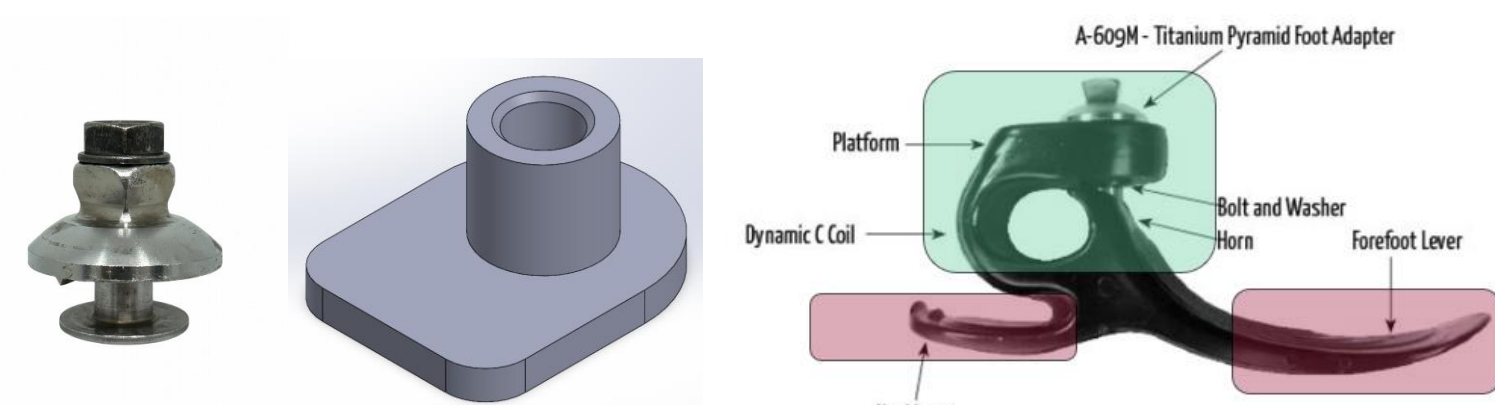
PROJECT BACKGROUND

In collaboration with DOI Ortho-Innovativ, the objective of the *Niagara Foot Project* is to improve worldwide access to high-functioning prosthetic feet for low-resourced regions [1]. Primary target areas include communities in Central America and Africa, where affordable prosthetics are vital to enhancing quality of life.

The overarching goal is to produce a foot that meets or exceeds the performance of the Niagara Foot Model 2 while enabling high-volume global supply at 30% of the previous cost. The newly established M3V01 targets this goal by re-engineering hardware components to eliminate high-cost manufacturing and post-processing steps within the supply chain.

DESIGN PROCESS

Through studying the fabrication process, the team discovered that the top plate and C-spring region have the most opportunity for cost savings regarding material reduction and post-processing; therefore, the team pursued design iterations on a new adapter insert piece.



Using a stage-gate approach, an emphasis on fabricating a new adapter insert was pursued. To place the new insert, two primary methods were developed:

1. Slice the foot in half in the sagittal plane.
2. Pot the adapter insert within the top plate to simulate overmoulding with a material similar to Hytrel 8238.

The potting strategy was prioritized based on timeline constraints. To complete testing this semester, the team developed an apparatus that could be easily replicated by the client. Several 3D printed guides were also developed to assist with the accuracy of insert placement.

Various design iterations of the adapter insert were manufactured and put through mechanical testing.

IMPLEMENTATION

Upon manufacturing of the proposed adaptor, it was discovered to be too large to integrate with the pyramid adaptor correctly.

A new, modified piece was to be made. This delayed the timeline, causing the team to re-evaluate. The 'pocket' for the adaptor was now up to the team to create.

A cutting template was 3D printed to allow for a dovetail drill bit to add a taper. This ensures a locking mechanism in the Hytrel-Polyurethane interface. However, this taper was done incorrectly on the first trial.



The pocket was created with the help of the cutting template, then a polyurethane cast was created using two components:

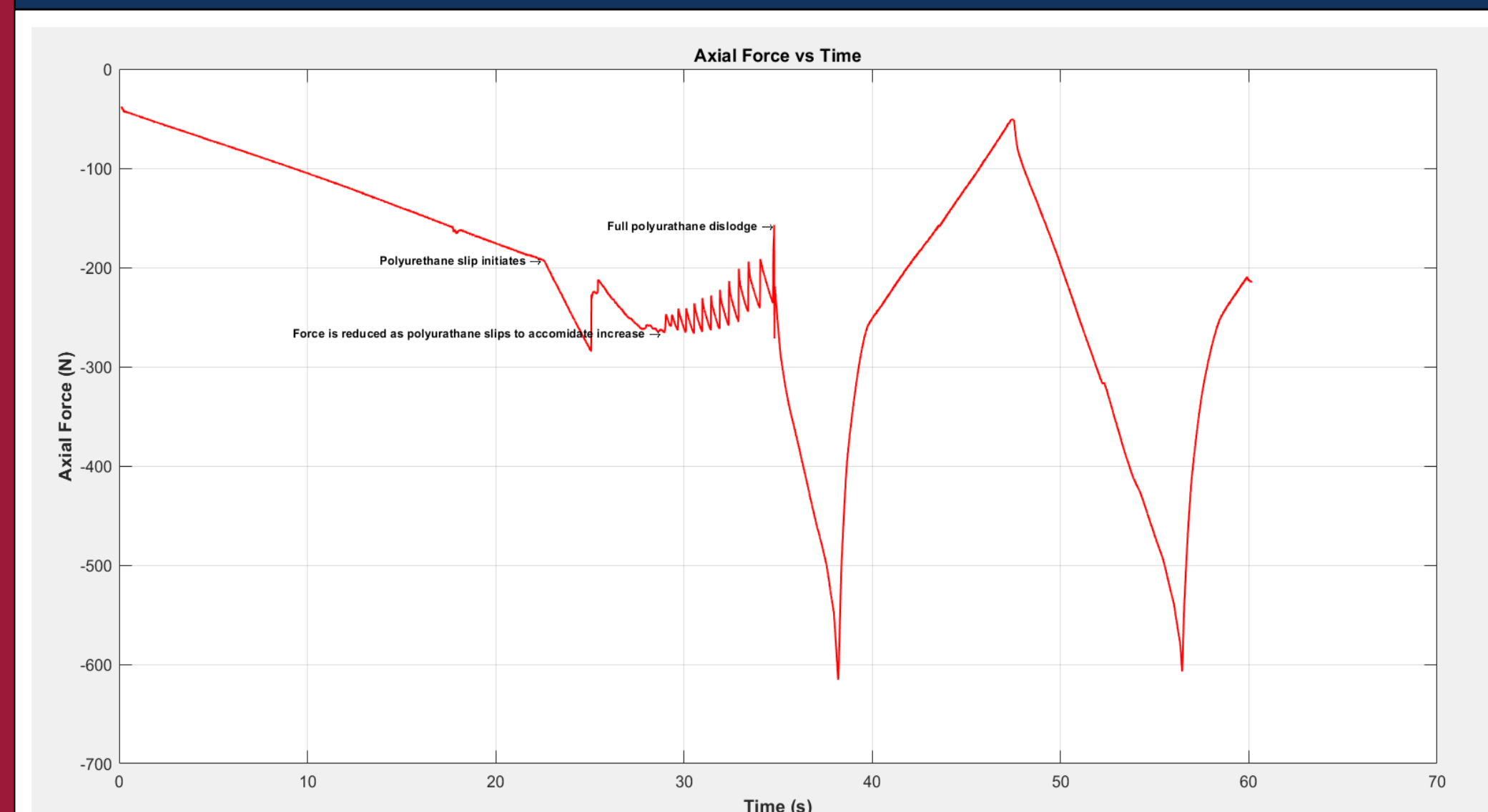
- ISO 380 A (liquid base, left)
- AMN 540 B (solid activator, right)

These components were mixed (40 g base, 10 g activator)

After mixing, it was poured into the pocket with a mock adaptor piece and left to solidify for 20 hours before testing.



VERIFICATION



The overall testing of the Niagara Foot's potting method performed as anticipated.

- With the most recent test, the polyurethane pocket loosened and eventually dislodged because it lacked a taper or locking feature.
- The adaptor component showed no slippage or deformation, proving that overmoulding is a viable manufacturing strategy moving forward.

MTS displacement-controlled testing following Multipurpose Elite test plan:

1. Ramp to 50N, dwell for 10 s
2. Repeat 5 times:
 - a. Ramp to 600N or -50mm
 - b. Return to 50N



OUTSTANDING RISKS

Adhesive Degradation of the Bearing Plate

The polyurethane bond between the bearing and top plates may lack the durability of the current mechanical fastening.

Impact: Humidity and repetitive stress cause delamination or shifting. This compromises roll-over consistency, increases audible impact noise (squeaking), and accelerates wear on the horns and top plate.

Mitigation: Replace adhesive with a mechanical interlock or a moisture-resistant chemical bond validated by shear stress testing.

POLYURETHANE-HYTREL BOND FAILURE

The interface between the polyurethane potting and the Hytrel 8238 cavity lacks sufficient chemical adhesion.

Impact: Strength testing confirms the weak bond between the polyurethane and Hytrel breaks immediately. This immediate point of failure allows the insert to exit the cavity, rendering the prosthetic unsafe and non-functional.

Mitigation: Machine internal ridges into the cavity walls to provide a mechanical lock and install a transverse side pin for redundant security. Overmoulding remains the definitive long-term solution.

NEXT STEPS

DOI-Ortho Innovativ will implement an inserted adapter design, scaling from prototype potting to automated overmoulding.

- **Hybrid Geometry Insert:** Adopt a circular-to-rectangular profile to optimize the area-to-force ratio. This neutralizes torsional stress and distributes gait loads over a larger footprint.
- **Manufacturing Optimization:** Replace manual potting with custom overmoulding tools to decrease labour, cut cycle times, and lower unit costs at scale.
- **Fatigue Validation:** Execute extended cyclic and field tests to map the fatigue life of the new geometry and confirm structural integrity.

REFERENCES

[1] J. T. Bryant, "Niagara Foot M2 V22 Post-Processing," internal company background document, Jun. 21, 2017.

Acknowledgements:

- Faculty Advisor, Dr. Tim Bryant
- DOI Ortho-Innovativ, Karlsberger Str. 3. 87471 Durach. Germany

